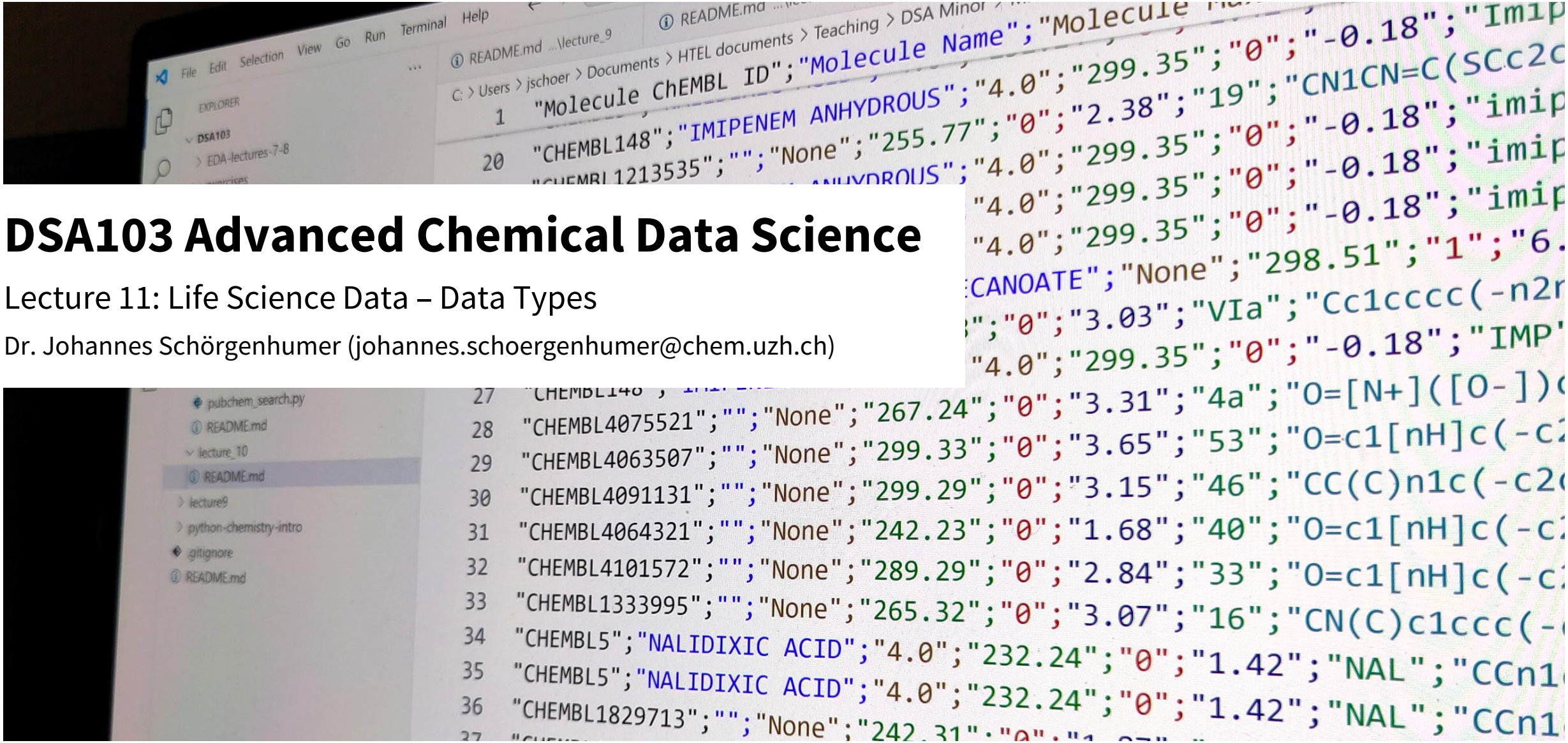
Universität Zürich[UZH]

Department of Chemistry



DSA103 Advanced Chemical Data Science

Lecture 11: Life Science Data – Data Types

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Content overview

Preface

Data Types

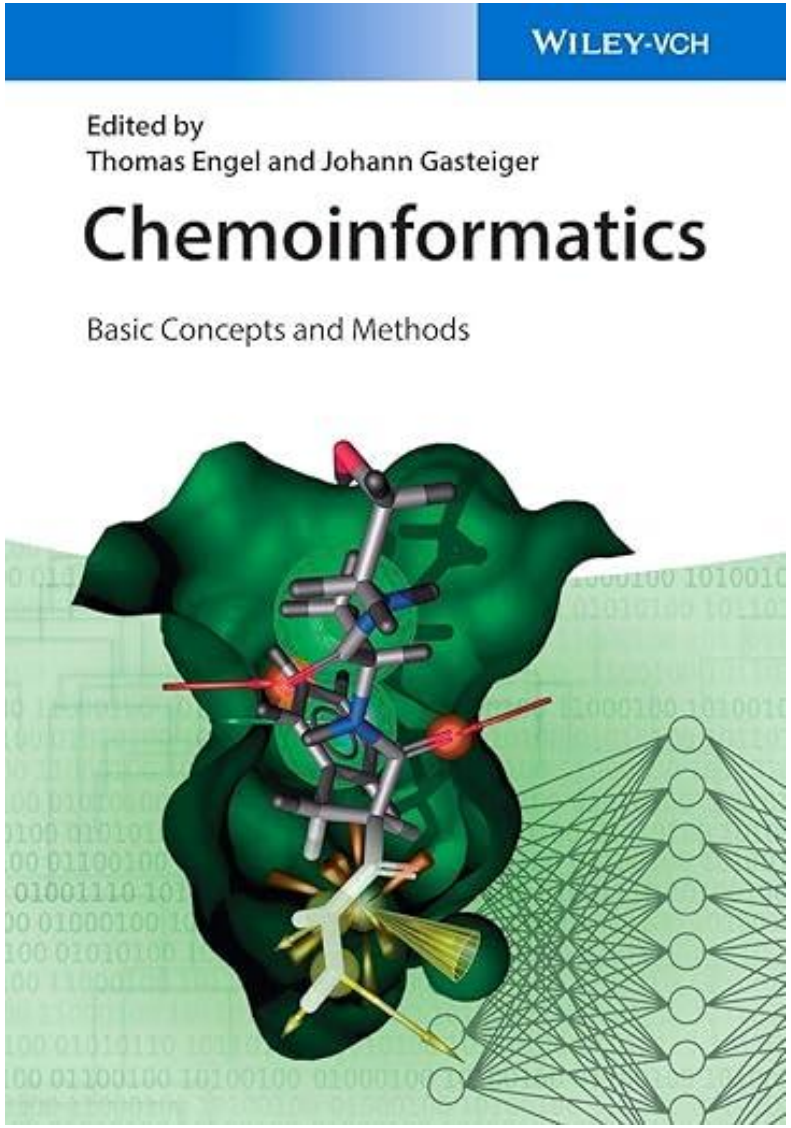
Introduction to molecular representations

Outline

Life Science Data is complex!

- High-dimensionality
- Heterogeneous
- Temporal and spatial complexity
- Noise
- Multiscale: Atoms – Macromolecules – Cells
- Reproducibility
- Standardization
- …

Field dependent specialities: Bioinformatics, Quantitative Biomedicine, Neuroinformatics, …



Engel, T., Gasteiger, J., *Cheminformatics* (1st ed.), 2018, Wiley-VCH

Main data types in the life sciences

Scientific data types ≠ python data types!

A lot of data in the life sciences is encoded in complex data type “structures” (arrays, graphs, stacks, lists, …)

What real information is encoded?

- Numerical data
- Textual / categorical data —> ie Experimental conditions
- Metadata & Experimental Conditions
- Sequence data
- Biological & functional data
- Computational data
- Spectral data
- Hybrid & Multimodal Data
- Structural data

Numerical data

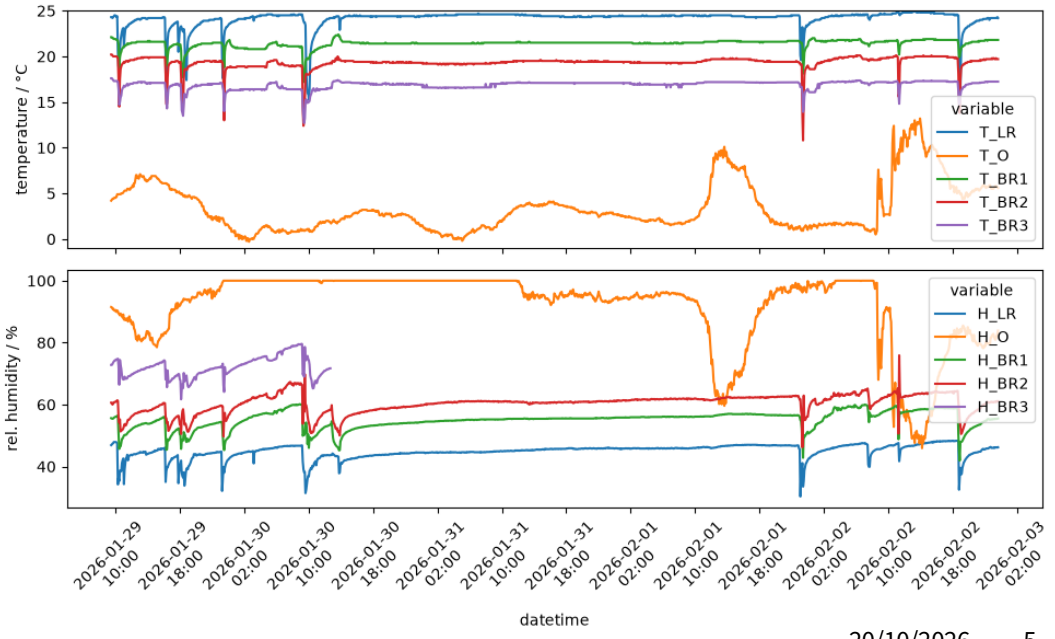
Most fundamental and widely used. Some important characteristics:

- Quantitative nature: numbers for comparison, calculations, etc.
- Complexity: mostly associated to units, may span wide range of orders of magnitude
- Multi-variance: Many values depend on other variables
- Significance level: single-values vs. averages. Statistics!
- Graphical impact: THE data type to plot and visualize!

Numerical data can be:

- **Deterministic**: Fixed values based e.g. on molecular composition (Molecular weight, number of heavy atoms)
- **Measured**: Experimental results, may have inherent variance (measurement error, different conditions, …)
- **Predicted**: Calculated by a certain model (value depends on model, method, version, …)
- Null **/ NaN**

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```



Tomczak J. (2018) *Data Types* In Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 156-168), Wiley-VCH.

Textual / categorial data

Non-numeric but structured data:

- Compound names, IDs
- 1D molecular structure encoding: SMILES, InChI
- Units
- Annotations and descriptions

```
molecule_chembl_id,canonical_smiles,standard_type,standard_relation,standard_value,stanc
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Tomczak J. (2018) *Data Types* In Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 156-168), Wiley-VCH.

Metadata & Experimental Conditions

Essential contextual information to describe the experiment:

- Numerical data : Temperature, pressure, solvent, pH, instrument parameters, calibration data
- Sample preparation protocols, time points, replicates, test types, …
- Materials and equipment used, reagent sources and preparation, …
- Provenance and versioning, especially for automated or high-throughput experiments

Often unstructured, textual data, however, there are developments in this direction, e.g.

- Open Reaction Database
- χDL by Cronin group

https://docs.open-reaction-database.org/en/latest/

Rauschen, M. Guy, J. E. Hein, L. Cronin, *Nat. Synth.*, **2024**, https://doi.org/10.1038/s44160-023-00473-6

1. General Information

NMR spectra were recorded on AV2-400 or AV2-500 MHz Bruker spectrometers at 298 K. FT-NMR spectrometer (^{1}H, 500 MHz; ^{13}C, 126 MHz; ^{31}P, 162 MHz; ^{19}F, 376 MHz; ^{11}B, 161 MHz). Chemical shifts δ are given in ppm. The spectra were calibrated using the residual chloroform signal (7.26 ppm for ^{1}H NMR and 77.16 ppm for ^{13}C NMR) and the residual dichloromethane signal (5.32 ppm for ^{1}H NMR and 54.84 ppm, for ^{13}C NMR). Multiplicities are abbreviated as follows: singlet (s), doublet (d), triplet (t), quartet (q), doublet-doublet (dd), multiplet (m), and overlap (ov). Infrared spectra were recorded on a JASCO FT/IR-4100 spectrometer. High-resolution electrospray ionization and electronic impact mass spectrometry were performed on a Finnigan MAT 900 (Thermo Finnigan, San Jose, CA; USA) double focusing magnetic sector mass spectrometer. Ten spectra were acquired. A mass accuracy ≤ 2 ppm was obtained in the peak matching acquisition mode by using a solution containing 2 <lPEG200, 2 <l PPG450, and 1.5 mg NaOAc (all obtained from Sigma-Aldrich, CH-Buchs) dissolved in 100 mL MeOH (HPLC Supra grade, Scharlau, E-Barcelona) as internal standard. Single-crystal X-ray diffraction data were collected at 183(1) K on a Rigaku OD XtaLAB Synergy, Dualflex, Pilatus 200K diffractometer using a single wavelength X-ray source (Mo Kαradiation: λ= 0.71073 Å) from a micro-focus sealed X ray tube and an Oxford liquid-nitrogen Cryostream cooler. The

χDL #1.1, executed with Graph #1.1

1) Add 2,4,6-trimethylaniline (2.75 mL, 19.6 mmol, 2.06 equiv.) directly to reactor_1 at default speed with stirring at 300 RPM.
2) Add THF (1.0 mL) directly to reactor_1 at default speed with stirring at 300 RPM.
   Adding 1 mL THF after reagent addition pushes residual reagent in the line between the Chemputer and reactor flask into the reactor.
3) Clean backbone with THF (3 × 3 mL).
   Cleaning the backbone here removes residual amine, preventing the glyoxal and amine from reacting in the lines and creating a solid and subsequent clog in the Chemputer.
4) Add glyoxal (40% wt. in H_2O) (1.09 mL, 9.50 mmol, 1.0 equiv.) directly to reactor_1 at default speed with stirring at 400 RPM.
5) Add THF (1.0 mL) directly to reactor_1 at default speed with stirring at 400 RPM.
   Adding 1 mL THF after reagent addition pushes residual reagent in the line between the Chemputer and reactor flask into the reactor.
6) Wait for 30 min.
7) Add THF (5.0 mL) directly to reactor_1 at default speed with stirring at 400 RPM.
8) Heat/Chill reactor_1 to 70 °C with stirring at 400 RPM.
9) Wait for 3 hours.

Sequence data

Mostly relevant to biochemistry and chemical biology:

- DNA/RNA sequences: Nucleotide strings (FASTA, GenBank)
- Protein sequences: Amino acid strings, annotated with modifications or mutations (pdb, swissprot, fasta)
- Peptidomimetic or synthetic sequence analogues: Noncanonical residue representations

```
>P01013 GENE X PROTEIN (OVALBUMIN-RELATED)
QIKDLLVSSSTDLDTTLVLVNAIYFKGMWKTAFNAEDTREMPFHVTKQESKPVQMMCMNNSFNVATLPAE
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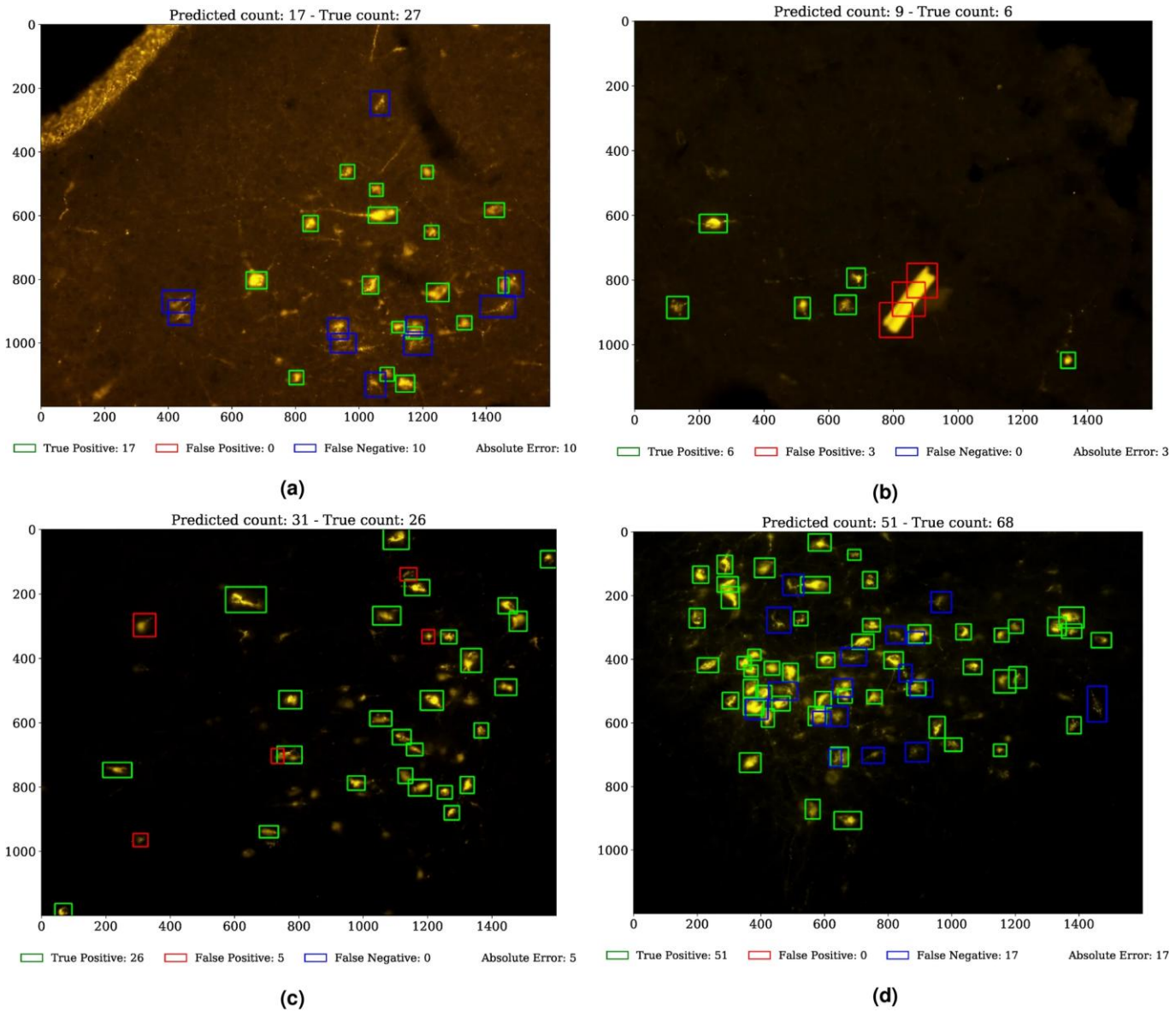
- Easy processing / computational decoding: string format

https://blast.ncbi.nlm.nih.gov/doc/blast-topics/

Biological & functional data

Describing how molecules behave in biological systems, e.g.:

- Enzyme kinetics: Michaelis–Menten plots, inhibition constants, turnover numbers
- Cellular assays: Dose–response curves, IC_{50}/EC_{50} values, viability percentages
- Omics data: Proteomics, metabolomics, transcriptomics — often tabular and high-dimensional
- Imaging data: Fluorescence microscopy, FRET, flow cytometry (FCS files), confocal microscopy images

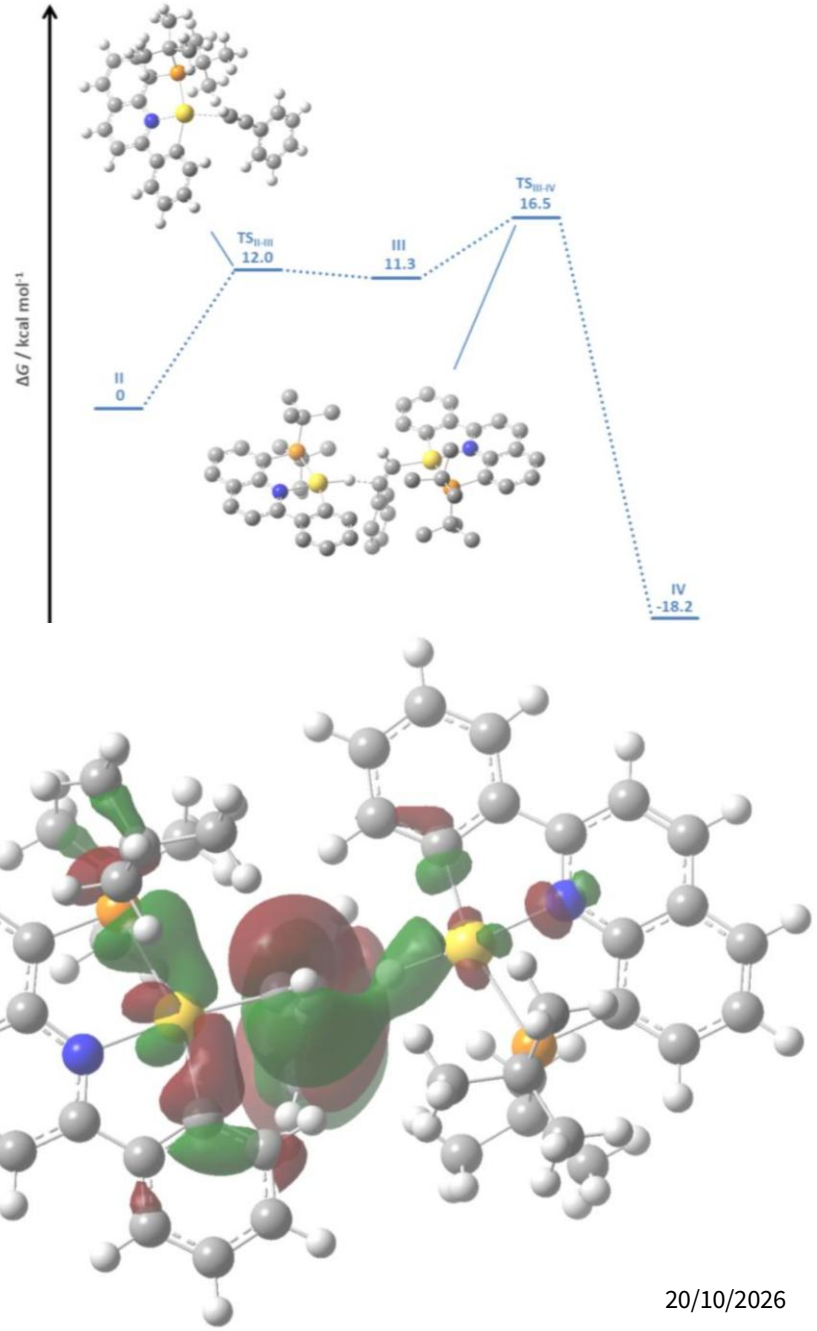


R. Morelli, L. Clissa, R. Amici, *et al., Sci Rep* **2021**, *11*, 22920, https://doi.org/10.1038/s41598-021-01929-5

Computational data

Generated in simulations, QM calculations, etc.:

- Quantum chemical calculations: Energies, orbitals, electron densities, potential energy surfaces
- Molecular dynamics: Time-resolved trajectories, RMSD/RMSF, energy profiles
- Docking results: Binding poses, scores, predicted affinities
- Cheminformatics descriptors: Molecular fingerprints, topological indices, physical properties (e.g. logP, dipole momentum, polar surface area)



J. Martín, J. Schörgenhumer, C. Nevado, *JACS Au* **2025**, *5*(3), 1439–1447, https://doi.org/10.1021/jacsau.5c00056

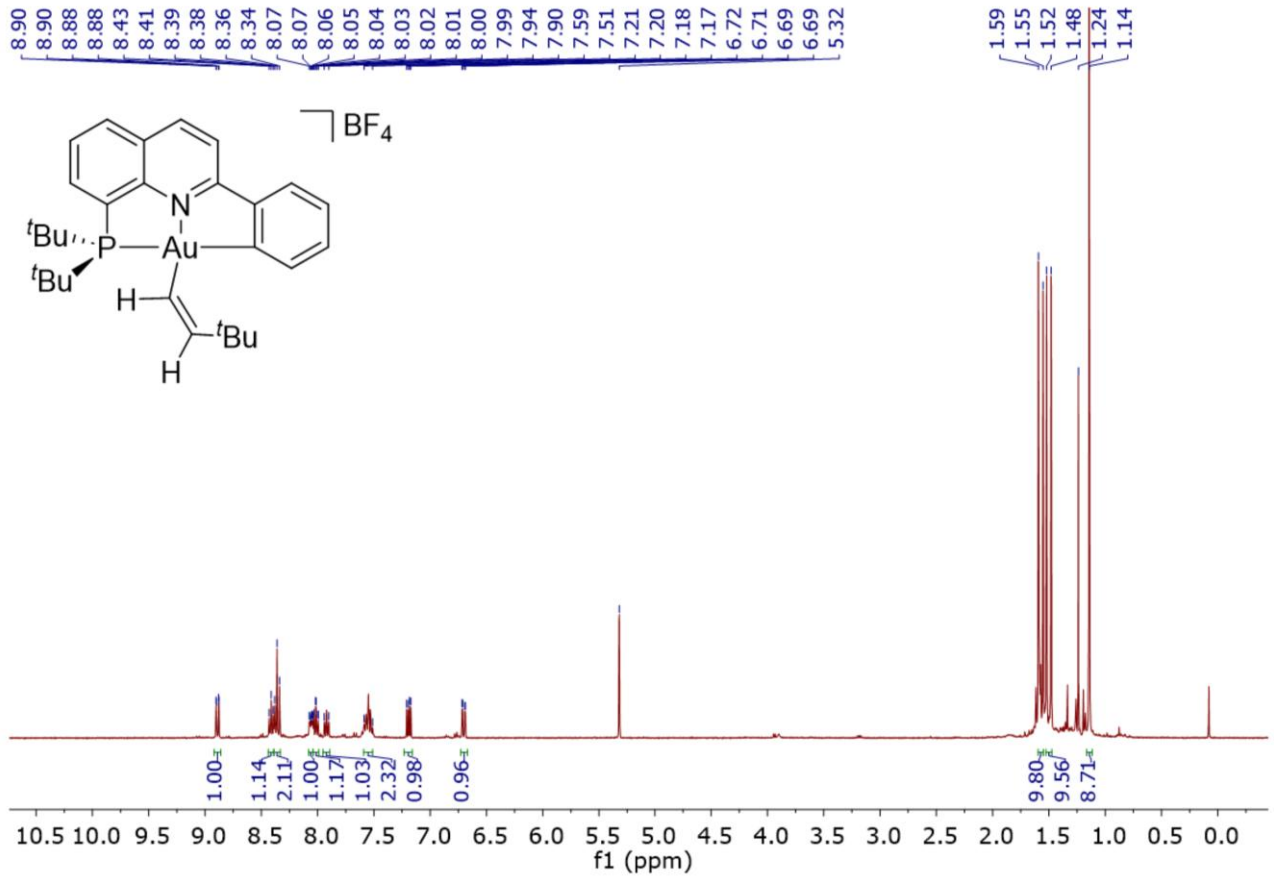
Spectral Data

Quite diverse data from different experimental setups, e.g. IR, NMR, MS, UV-vis, …

All have in common that the results are linked to the molecular structure!

Various representations, e.g.

- Fully digitalised spectrum
- Sampled spectrum
- Peak table representation
- Binary string



Tomczak J. (2018) *Data Types* In Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 156-168), Wiley-VCH.
J. Martín, J. Schörgenhumer, C. Nevado, *JACS Au* **2025**, *5*(3), 1439–1447, https://doi.org/10.1021/jacsau.5c00056

Hybrid / Multimodal Data

- Multimodal data refers to datasets that combine multiple types (or "**modes**") of information about the same system, sample, or phenomenon
- Each **mode** captures a different **aspect** of the system — for example, its structure, composition, dynamics, or biological effect.
- Integrating more aspects: more complete and nuanced understanding than with any single data type alone
- Quite common in life sciences (complex questions…)

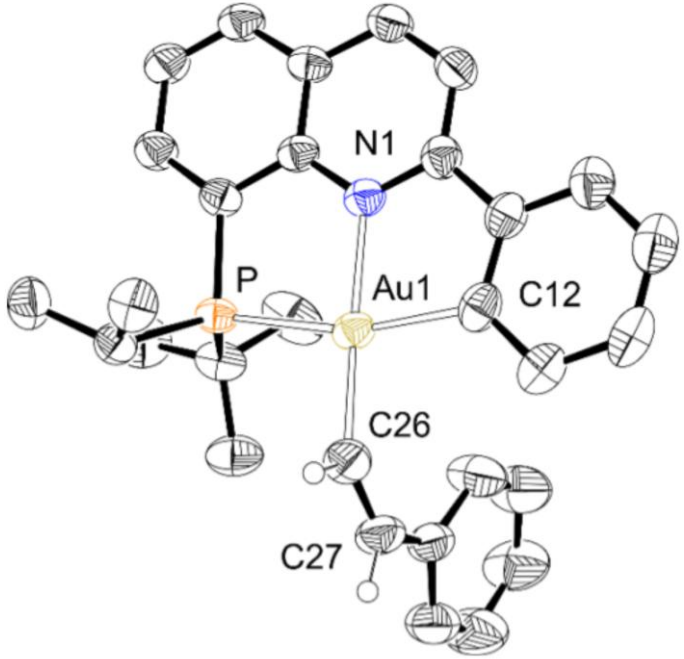
Examples:

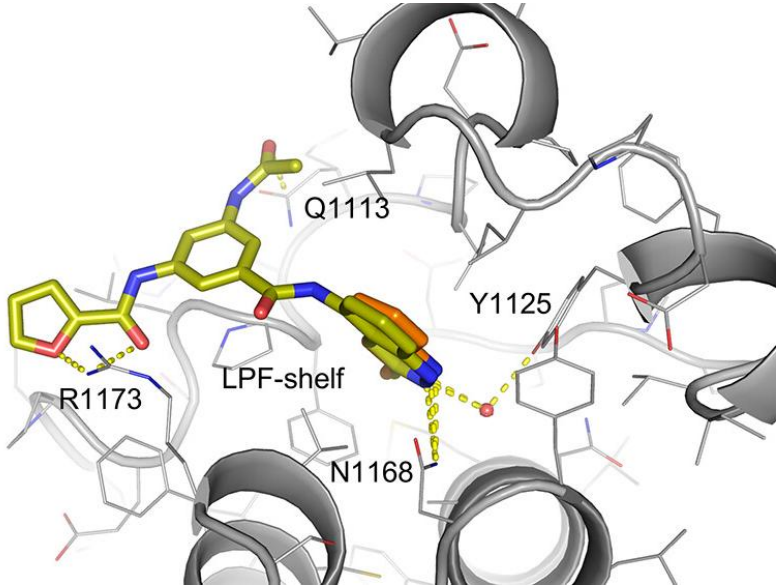
- Drug discovery: Structural data (protein–ligand complex) + screening data (IC_{50} values) + transcriptomics (gene expression changes)
- Catalyst discovery: Molecular structure (Xray, NMR, MS) + reaction data (Yields) + kinetics (reaction rates)

Structural data

Information about **molecular geometry and spatial arrangement**:

- Molecular structures:
  - Atom coordinates (e.g., from X-ray crystallography, cryo-EM, NMR)
  - File formats: .pdb, .mol2, .xyz, .cif
- Electron density maps / volumetric data:
  - 3D grids of electron density (e.g., from cryo-EM or crystallography)
- Computational models:
  - Molecular dynamics trajectories, energy minimization outputs, docking poses





J. Martín, J. Schörgenhumer, C. Nevado, *JACS Au* **2025**, *5*(3), 1439–1447, https://doi.org/10.1021/jacsau.5c00056
K. A. Gosselé, I. Latino, E. Laul, M. S. Kirillova, V. Pascanu, E. Carloni, R. K. Bedi, C. Pizzichetti, A. Caflisch, S. F. González, C. Nevado, *JACS Au* **2025**, *5*(6), 2491–2499, https://doi.org/10.1021/jacsau.5c00085

Acitivity: Identify datatypes in complex life science data

1) Identify data types in the following data entries:

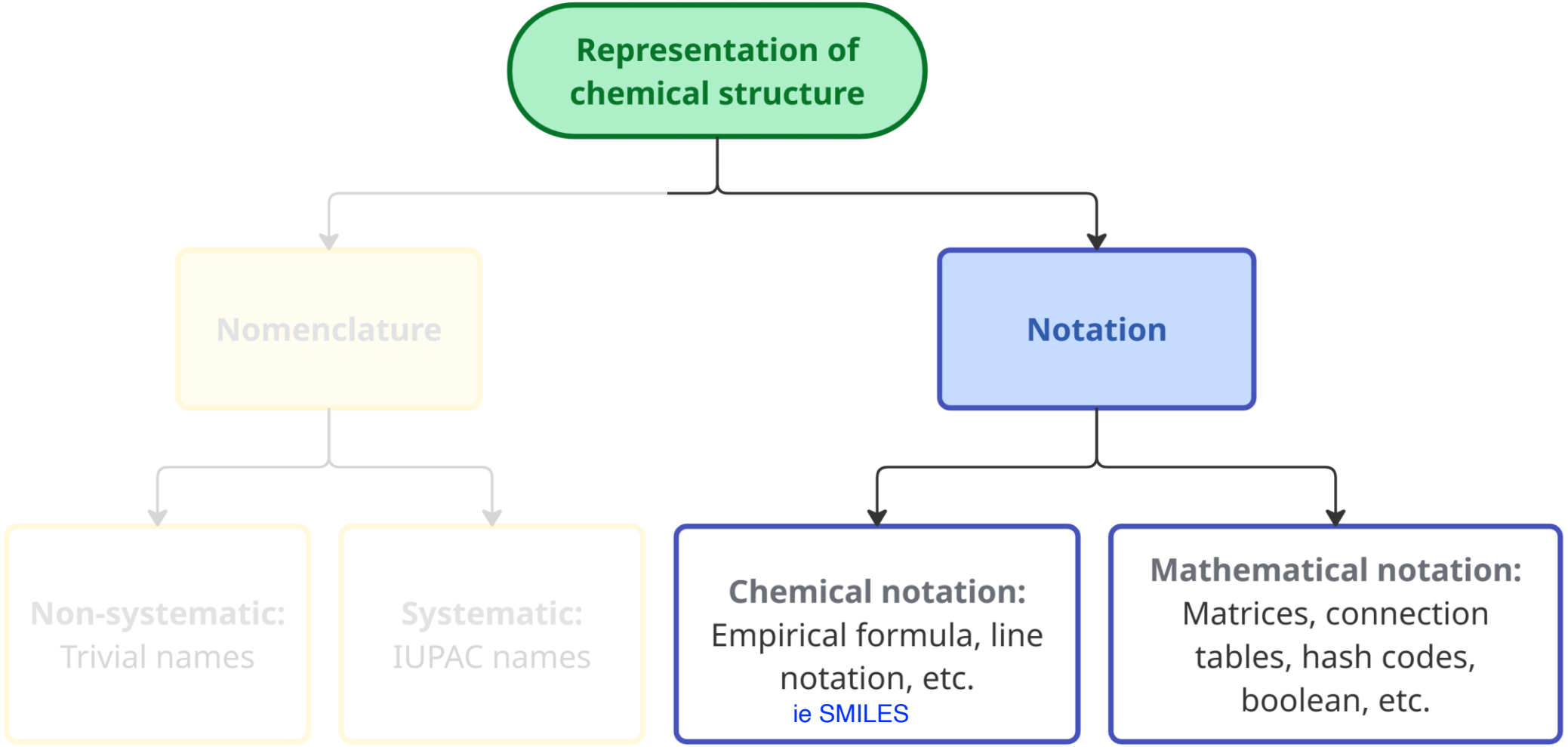
- Computational output file Numerical, Structural Metadata, numerical, structural
- ELN / ELN export Metadata, Sequence data (SMILES) Textual (most of it)
- ORD entry Experimental conditions, essntual contextual information Numbers and text, metadata

2) Imagine a task specific to your domain (e.g. reaction setup, cytotox assay).

- Outline the process in simple steps!
- Which data types do you record at each of them?

Catalyst (performance) screening
- Experimental conditions
- Analytical data in numerical form & as plots

Introduction to encoding molecules



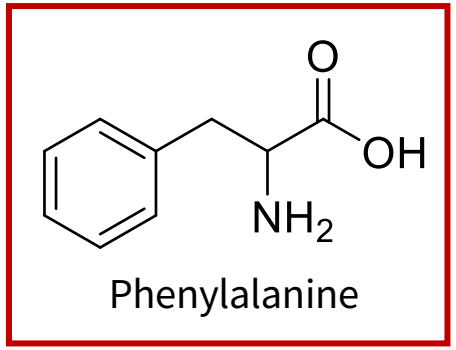
Engel T. (2018) *Principles of Molecular Representations* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-41), Wiley-VCH.

Encoding molecular structures

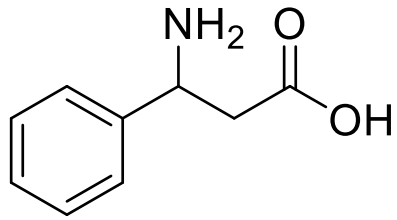
1D chemical notations:

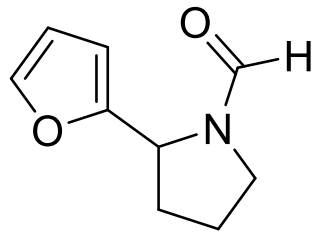
Empirical formula / “Sum formula”: e.g. $C_9H_{11}NO_2$

Not unambiguous: Cant derive structure from that



Phenylalanine





SMILES: NC(Cc1ccccc1)C(O)=O
Or: C1=CC=C(C=C1)C[C@@H](C(=O)O)N
Or: NC(C(=O)O)(CC1=CC=CC=C1)

A bit better, gives idea but different SMILES give the same picture

SLN (SYBYL Line Notation): C[1]H:CH:CH:CH:CH:C(:@1)CH2CH(NH2)C(=O)OH

InChI (IUPAC International Chemical Identifier): 1S/C9H11NO2/c10-8(9(11)12)6-7-4-2-1-3-5-7/h1-5,8H,6,10H2,(H,11,12)/t8-/m0/s1

- Normalized (aromaticity, tautomers)
- Canonicalized (unambigous labelling)
- Serialized (linear string as output)

Chemical formula | Connection sublayer | H atom sublayer | stereochemistry

InChIKey: Compressed, not human-interpretable 27 character string computed from InChI for facilitated search, e.g. Phenylalanine = COLNVLDHVKWLRT-QMMMGPOBSA-N

Engel T. (2018) *Principles of Molecular Representations* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-41), Wiley-VCH.

Encoding molecular structures - SMILES

“Simplified Molecular Input Line Entry System”, 1986 by D. Weininger

- Graph-based
- Highly compressed
- Still human-readable
- Dependent on canonicalization algorithm (!)

Different modifications for different fields of application (XSMILES, SMIRKS, CHUCKLES, …), e.g. polymers, reactions, databases

Various algorithms for generating canonical SMILES, e.g. by Daylight or OpenEye; also **implemented in RDKit**

SMARTS (Smiles Arbitrary Target Specification): Smiles, but with…

- Wildcards: “*” (any atoms), “~” (any bonds), “@” (any ring bond)
- Lists of elements: [N,C,O]
- And logic operators: “!” (NOT), “&“,” (AND), “,” (OR)

Engel T. (2018) *Computer Processing of Chemical Structure Information* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 43-119), Wiley-VCH.
https://deepwiki.com/rdkit/rdkit/3.1-smiles-and-smarts-processing

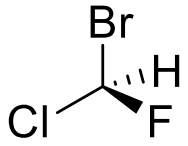
Encoding molecular structures - SMILES

Examples SMILES:
Valeriansäure: CCCCC(=O)O
Pyridin: n1ccccc1
Chinolin: n1cccc2c1cccc2

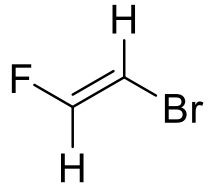
Explore SMILES Code

Basic rules:

1) Atoms = atomic symbols (Ambiguous two-letter symbols have to be written in brackets, e.g. [Nb] != NB)
2) Hydrogen atoms omitted (saturate free “valences”; valences standardized for atoms)
3) Neighbouring atoms stand next to each other
4) Only double (=) and triple bonds (#) are indicated explicitly, aromatic connections (:) are deprecated and should be substituted by use of lower-case letters
5) Branches are to be written in parentheses and can be nested
6) Rings are described by digits indicating the connecting atoms (e.g. C1CCCC1 = cyclopentane)
7) Charges of an atom are to be indicated in brackets, e.g. [Cu2+]
8) Separate molecules are separated by a period “.”
9) Stereochemical information is encoded by using @ (counterclockwise **reading order of connected atoms in smiles, does not correlate with R/S nomenclature!!**) and @@ (clockwise), isomers on double bonds use “/” and “\” to indicate the relative direction.



Br[C@H](Cl)F or Br[C@@H]F(Cl)



F/C=C/Br or F\C=C\Br

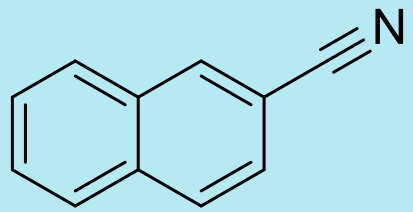
Engel T. (2018) *Computer Processing of Chemical Structure Information* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 43-119), Wiley-VCH.

Acitivity: SMILES practice

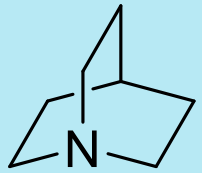
1) What molecule is encoded in the string CSCC[C@@H](C(=O)O)N ?

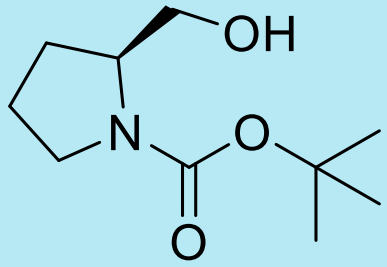
L-Methionin

2) Derive the SMILES string for the following molecules:



C1=CC=C2C=C(C=CC2=C1)C#N





3) Use RDKit to visualise your SMILES codes. If matching the structures, convert them back into Smiles.